SASBPageRAG at the NTCIR-19 RegCom Task: Retrieval-Augmented Few-Shot Verification for Single-Page ESG Compliance

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Abstract

This paper reports SASBPageRAG, a compact retrieval-augmented system for the NTCIR-19 RegCom single-page metric verification task. Given a SASB metric and an ESG report page, the system predicts whether the page addresses the metric and, if so, whether the disclosure category and unit of measure comply with the SASB requirement. Instead of fine-tuning a language model, we use the training data as a retrieval memory. TF-IDF retrieval selects semantically similar and label-balanced few-shot examples, which are combined with SASB metadata and extracted PDF page text in a structured verifier prompt. On 793 non-Chinese test rows, Gemini 2.5 Pro and GLM-5.2 obtain Macro F1 scores of 0.6140 and 0.5963, respectively, compared with 0.4489 for a MiniLM embedding classifier. A completed GPT-5.4-mini page-image run obtains 0.5748 Macro F1. We further compare our text-based verifier with two other RegCom submissions. Across the independently reported experiments, metric-prioritized demonstrations and page images emerge as promising complementary signals, while all three studies identify partial disclosure as a central difficulty. Because the submissions differ in language coverage, retained instances, data cleaning, and input modality, we treat their scores as contextual rather than as a direct ranking. We additionally report a preliminary, diagnostic-only retrieval extension toward the official Subtask 1 (evidence-page retrieval) setting; these are explicitly not official Task 1 results and are included to document reproducible work in progress.

Keywords

ESG compliance, SASB, retrieval-augmented generation, few-shot learning, large language models, multilingual NLP

Team Name

IMNTPU

Subtasks

RegCom Task 1 - Subtask 2: Single-Page Metric Verification. This report focuses on the non-Chinese subset: English, French, Japanese, Korean, and Thai. Section 4.7 additionally reports a preliminary, diagnostic-only extension toward Subtask 1 (multilingual evidence-page retrieval); it is not an official Task 1 result. The task description defines Subtask 1 more broadly as full-report compliance matching (page localization followed by category/unit verification); our reported Task 1 numbers cover only the page-localization stage.

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1 Introduction

Sustainability reporting standards require firms to disclose environmental, social, and governance (ESG) information according to specific metric definitions, disclosure categories, and units of measure. The Sustainability Accounting Standards Board (SASB) standards are representative of this metric-oriented style: a disclosure is not only about a topic, but also about whether it satisfies a precise metric, category, and unit requirement [1]. ESG reports are long, multilingual, and often semi-structured, so page-level compliance checking is costly for regulators, assurance providers, and analysts.

The NTCIR-19 RegCom task formulates this challenge as multinational, multilingual, and multi-industry regulatory compliance verification for ESG reports [2, 3]. In Subtask 2, each instance contains a SASB metric row and a single page from an ESG report. A system must predict three fields: `pred_label`, `category_match`, and `unit_match`. The main label has three values: `yes`, `yes but not complete`, and `no`. This makes the task harder than topical relevance classification. A page may discuss the same ESG topic yet fail to disclose the required quantity, omit sub-items, use a proxy, or report an incompatible unit.

This paper presents SASBPageRAG, a lightweight retrieval -augmented verifier for single-page SASB metric verification. The system uses the labeled training rows as a retrieval memory and asks one LLM verifier to make a structured page-level judgement. The design follows the intuition of few-shot prompting [4] and retrieval-augmented generation [5]: rather than updating model parameters, the system supplies task-relevant examples at inference time. This design allows us to compare strong closed LLMs with a fast open-source embedding classifier under the same data split and output schema.

The contribution of this report is fourfold. First, we describe a retrieval-augmented few-shot prompting strategy for SASB page-level verification. Second, we compare Gemini 2.5 Pro, GLM-5.2, and a MiniLM logistic-regression baseline on the same 793 non-Chinese test rows. Third, we analyze class-level and language-level behavior, showing that the main remaining challenge is the boundary around partial disclosure. Fourth, we place these findings in the context of the Cierpa and CSCU submissions, comparing retrieval design, evidence modality, few-shot selection, evaluation scope, and reported limitations without treating non-aligned scores as a leaderboard.

Beyond Subtask 2, Section 4.7 reports a preliminary, diagnostic retrieval extension toward the official Subtask 1 (evidence-page retrieval) setting, built from already-cached local runs. These numbers are not official Task 1 results: they are included to document reproducible, in-progress work and to state plainly which prerequisites – an official evaluator and a confirmed gold page-set contract – remain unresolved.

2 Related Work

2.1 ESG Disclosure Analysis and Multilingual Verification

ESG disclosure analysis focuses on evaluating the completeness and reliability of environmental, social, and governance information in corporate sustainability reports [6]. Since these reports may be written in different languages and vary across regions, industries, and reporting formats, multilingual verification has become an important issue in sustainability analysis [7]. Researchers need to determine whether the disclosed content satisfies specific reporting standards and whether the meaning remains consistent across languages. This study focuses on the relationship between ESG reports and SASB metrics, using model-based methods to classify disclosures as complete, partially complete, or not disclosed.

Prior ESG NLP research has shown that domain-specific language models and evidence extraction are useful for sustainability and financial text. ClimateBERT adapts transformer language modeling to climate-related corporate and news text [8], while Kannan and Seki extract textual evidence for ESG scores from financial reports [9]. These studies motivate the use of report evidence rather than only document-level labels, but they do not directly address SASB metric-level compliance decisions.

2.2 Multilingual Promise and Compliance Verification

Multilingual promise and compliance verification has become an important research direction in ESG text analysis. Corporate sustainability reports are often published in different languages, and the same commitment or disclosure may vary across languages, industries, and reporting formats. Therefore, models need to determine whether the commitment, supporting evidence, and standard requirements are semantically consistent. Recent work such as ML-Promise introduces multilingual corporate promise verification for ESG reports and studies evidence assessment across languages [10]. SemEval-2025 Task 6 further frames corporate promise verification as a multilingual and multi-industry task involving commitment detection and evidence consistency [11]. In contrast, NTCIR-19 RegCom focuses more on compliance verification by asking whether a single report page satisfies a specific SASB metric, including category and unit compatibility [3]. Such tasks therefore emphasize not only whether a topic is mentioned, but also whether the disclosure actually meets the required standard.

2.3 Retrieval-Augmented and Embedding-Based Methods

Our method is also related to few-shot prompting, retrieval-augmented generation, classical term weighting, and sentence embeddings. Large language models can perform new tasks from a small number of demonstrations [4], and retrieval-augmented generation supplies task-relevant external evidence at inference time [5]. We use TF-IDF because it is a strong, transparent lexical retrieval baseline for term-weighted document matching [12]. Sentence-BERT and multilingual knowledge distillation provide efficient embedding baselines for cross-lingual semantic matching [13, 14]. SASBPageRAG combines these ideas by using retrieval to select few-shot cases for a structured single-page verifier.

2.4 Prompt-based Large Language Models for ESG Classification

Prompt-based Large Language Model classification uses natural language instructions to guide models in performing specific classification tasks without necessarily retraining model parameters. In ESG disclosure classification, disclosure standards, report segments, supporting evidence, and label definitions can be integrated into the prompt, allowing the model to judge whether a company's disclosure satisfies a specific metric. Compared with traditional classifiers, prompt-based methods are more suitable for long, unstructured reports and diverse semantic expressions. When combined with a few examples or retrieved similar cases, the model can better understand the differences among labels such as complete disclosure, partial disclosure, and no disclosure. Therefore, prompt-based LLM classification provides a flexible approach for automated ESG report analysis.

2.5 Comparison with Other RegCom Submissions

Two other submitted systems provide useful contrasts to our page-text RAG design. Cierpa decomposes full-report compliance checking into relevant-page retrieval and label inference [15]. Its retrieval pipeline combines BM25, text embeddings, VLM-generated descriptions, image embeddings, reciprocal-rank fusion, and VLM reranking. For Subtask 2, it compares zero-shot inference with few-shot and extended few-shot prompts, using either random or metric-prioritized example selection. This design tests both evidence localization and demonstration choice, but requires several embedding and generative components and incurs additional reranking cost.

CSCU instead emphasizes traceable evidence grounding [16]. Its Subtask 1 system uses table-of-contents-first section localization with a full-document fallback and a VLM-guided candidate verifier. For Subtask 2, it directly compares text-only, image-only, and text-plus-image page evidence. This design isolates the value of page modality more directly than our experiments, although its strongest retrieval variant jointly changes section selection and final page verification, so the gain cannot be attributed to a single component. In contrast, SASBPageRAG assumes the task-provided page and concentrates on lightweight case retrieval and metric-level compliance reasoning from extracted text.

A terminology note is necessary before comparing these systems further: "Task 1" and "Subtask 1" are not used consistently across the three reports. We follow the official convention, in which RegCom Task 1 contains Subtask 1 (evidence-page retrieval within a full report) and Subtask 2 (single-page verification, the setting of this paper). CSCU's usage matches this convention directly. Cierpa instead labels its full retrieval-plus-verification pipeline "Subtask 1" and treats page retrieval as a sub-problem within it, so a Cierpa

"Subtask 1" number is not the same quantity as a CSCU or official Subtask 1 number. Section 4.7 of this report uses the official, CSCU-aligned sense of the term.

Beyond the retrieval-versus-verifier framing already summarized above, the two submissions differ in ways that shape how their numbers should be read. Cierpa evaluates four retrieval signals individually before fusion: BM25 lexical matching (MRR 0.231), text embeddings (MRR 0.368), VLM-generated element descriptions (MRR 0.436), and page-image embeddings (MRR 0.511, the strongest single signal), with an untuned BM25+VTE+IE union (MRR 0.527) outperforming its own weight-tuned fusion (MRR 0.497) – indicating that weight tuning on the training split did not generalize to the test split. Cierpa also audits the official RegCom release directly: across all six languages, 21.1%/16.5% (train/test) of relevant-page targets are non-unique for the same metric tuple, and 55.6%/57.2% of instances share a non-unique gold label at the PDF level, because a no label means the metric is absent from that specific page rather than from the whole report. Cierpa removes conflicting tuples and a page-numbering offset from its training split only (942 to 903 rows) and leaves test data unmodified. CSCU instead audits the official Subtask 1 split for document overlap and finds that all 62 official test documents are also present in the training split, so it explicitly reframes its own Subtask 1 results as seen-document, unseen-query evaluation rather than unseen-document generalization – a caveat that plausibly extends to the other two submissions as well, since all three draw on the same official release, though this has not been independently verified for the non-Chinese split used here. CSCU further reports that its VLM-Guided retrieval variant raises paired Subtask 1 macro F1 from 14.8 to 31.0 across 438 metric instances (95% bootstrap CI [11.0, 21.5] on the paired gain; Wilcoxon signed-rank $p < 10^{-6}$), while Near@1 – crediting a prediction one page from gold – rises from 10.4 to 46.6, roughly one and a half times the exact-match gain. Neither Cierpa nor SASBPageRAG reports a paired confidence interval or significance test for its headline comparison, which we note as a methodological gap relative to CSCU in Section 5.4.

3 Methods

Subtask 2 is Single-Page Metric Verification. Given a single SASB metric and a single page from an ESG report, the objective is to determine whether that page contains relevant information and, if so, whether the information complies with the specified SASB category and unit of measure. Each instance contains structured SASB metadata, extracted page text, and three prediction targets: `pred_label`, `category_match`, and `unit_match`. The main label is `yes`, `yes but not complete`, or `no`; if the prediction is `no`, the auxiliary fields are normalized to `N/A`. Because only one page is available, the task requires fine-grained local evidence matching rather than full-report compliance analysis.

The experiments use the non-Chinese subset of the task package, covering English, French, Japanese, Korean, and Thai. The retrieval and classifier training pool contains 753 labeled rows, and the evaluation set contains 793 rows. The training pool contains 295 `yes`, 127 `yes but not complete`, and 331 `no` rows, while the test set contains 315 `yes`, 138 `yes but not complete`, and 340 `no` rows.

3.1 Proposed Research Architecture

Figure 1 shows the proposed research architecture of SASBPageRAG. The workflow starts from training and development CSV files, SASB knowledge bases, and source PDFs. After preprocessing and evidence extraction, the system retrieves similar labeled cases, constructs a prompt with few-shot examples and JSON output constraints, applies LLM-based classification, and produces evaluation scores and prediction files.

3.2 Retrieval-Augmented Few-Shot Verifier

SASBPageRAG does not fine-tune the LLM. Instead, it treats the training CSV as a case library. For each test row, we build a retrieval document from the SASB topic, metric description, metric code, category, unit, key terms, `sasb_what_counts`, and the first 1,500 characters of extracted page text. A TF-IDF vectorizer with unigram and bigram features computes cosine similarity between the test row and all training rows [12].

The example selector retrieves three demonstrations. It first prioritizes rows with the same SASB metric code. Within that pool, it attempts to select the most similar example for each label: `yes`, `yes but not complete`, and `no`. If a label is missing for the same metric, the selector falls back to the global training pool. Remaining slots are filled by the most similar same-metric rows and then by the most similar global rows. This strategy is both similarity-based and label-balanced, reducing the chance that the prompt is dominated by the majority class.

The verifier prompt contains a compact system instruction, the retrieved demonstrations with gold outputs, the metric-specific label prior, the target SASB metadata, and the extracted PDF page text. The LLM is instructed to judge only the given page, not the full ESG report, and to return compact JSON containing `pred_label`, `category_match`, `unit_match`, short evidence, and a short reason. The prompt also states the boundary rules: relevant and complete disclosure is `yes`; partial, proxy-only, incomplete, or wrong-unit disclosure is `yes but not complete`; and semantically absent metric evidence is `no`.

3.3 Prompt Design

The implementation uses the same prompt structure for Gemini 2.5 Pro and GLM-5.2. The prompt is designed to make the model judge only the given page, calibrate its decision with retrieved examples, and return a valid task-format prediction. The following prompt template summarizes the main structure used in our system:

> *You are a SASB page-level verification assistant. Judge only the single PDF page provided. Return compact JSON only with the following fields:*
> `{"pred_label": "yes|yes but not complete|no",`
> `"category_match": "yes|no|N/A",`
> `"unit_match": "yes|no|N/A",`
> `"evidence": "short page evidence",`
> `"reason": "short reason"}`
> *Rules:*
> *1. Predict* `yes` *if the page contains relevant disclosure for the requested SASB metric.*
> *2. Predict* `yes but not complete` *if the page is relevant but partial, proxy-only,*
> *incomplete, or wrong-unit.*

Table 1: Design comparison across three RegCom submissions. “RAG examples” refers to retrieval of labeled demonstrations for the verifier, whereas “page retrieval” refers to localization within a full report.

System	Task focus	Page evidence	Page retrieval	RAG examples	Main trade-off
SASBPageRAG	Subtask 2	Extracted text	Task-provided page	TF-IDF, metric-first, label-balanced	Lightweight; loses layout
Cierpa	Subtasks 1–2	Text and page image	Lexical, semantic, visual, VLM reranking	Random or metric-prioritized	Broad signals; higher cost
CSCU	Subtasks 1–2	Text and/or page image	ToC-first, fallback, VLM guidance	Prompt-based verification	Traceable and multimodal; coupled changes

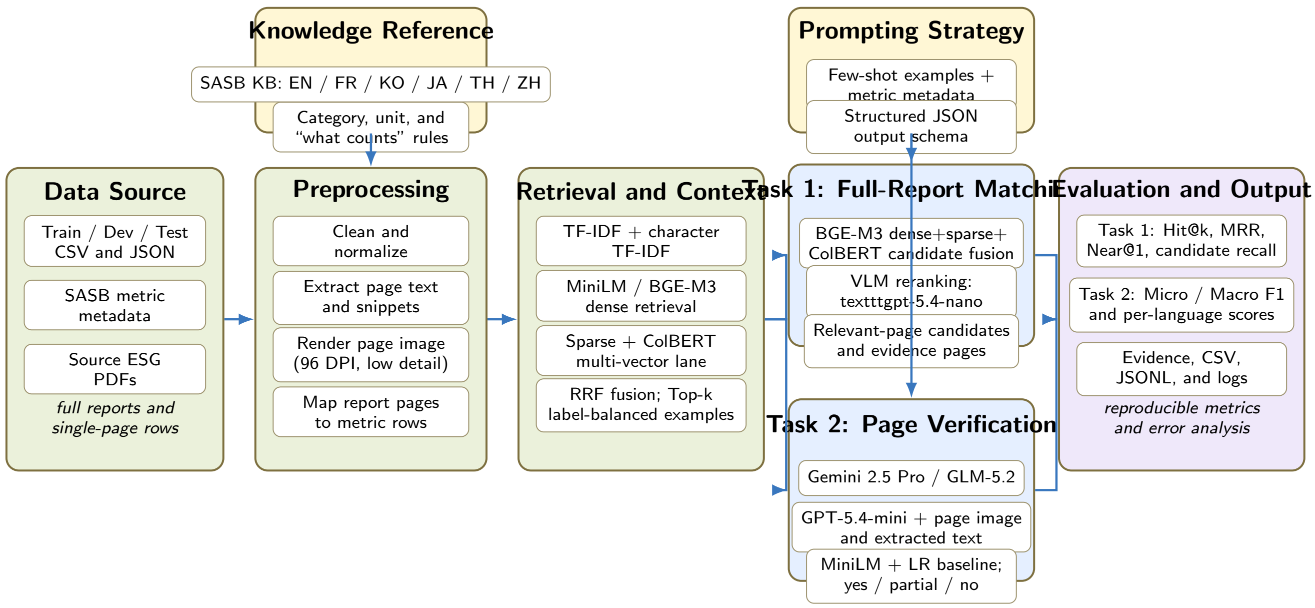

Figure 1: Unified research architecture for Task 1 full-report page matching and Task 2 single-page metric verification. Task 1 uses BGE-M3 multi-representation candidate generation with VLM reranking; Task 2 compares text LLMs, GPT-5.4-mini page-image verification, and a MiniLM baseline.

3. Predict `no` *if the exact metric is not semantically disclosed.*
4. If `pred_label` *is* `no`*, set* `category_match` *and* `unit_match` *to* `N/A`.
5. Use the retrieved few-shot examples and metric label distribution only as calibration. The target page evidence decides.
Retrieved few-shot examples:
Example 1–3: task metadata, evidence snippet, similarity score, same-metric flag, and gold output.
Now predict this test row:
Task: language, company/report id, file stem, page, metric code, topic, metric description, SASB category, unit, key terms, and what-counts guidance.
PDF page text: [extracted page text, truncated to 5,000 characters]
Return JSON only.

In the actual implementation, each retrieved example includes a page evidence snippet of up to 1,200 characters, its TF-IDF similarity score, whether it shares the same metric code as the target row, and the gold output. The prompt also includes the training label distribution for the same metric as a calibration prior. After generation, invalid or missing auxiliary labels are normalized by deterministic post-processing. For discussion-and-analysis metrics, relevant narrative evidence is treated as category- and unit-compatible, and zero-event or no-incident statements can satisfy event-count metrics when the metric asks for occurrences or counts.

3.4 Compared Systems

We evaluate two LLM configurations using the same retrieval and prompt design. The first uses Gemini 2.5 Pro because the Gemini 2.5 model family is designed for advanced reasoning, multimodality, and long-context processing [17]. These capabilities are relevant to fine-grained page-level verification, where the model must compare local report evidence with a structured metric definition. The second uses GLM-5.2 through the GLM API as an alternative commercial LLM from the GLM family. Public GLM-family reports emphasize strong instruction following, long-context ability, and multilingual pretraining [18]. Both systems receive the same structured task fields, PDF page text, and three few-shot examples.

As a multimodal follow-up, we also run GPT-5.4-mini through the OpenAI Responses API. The target PDF page is rendered at 96 DPI and supplied as a low-detail image together with the extracted page text; the retrieved demonstrations remain text-only. This run is evaluated on the same 793 non-Chinese rows and is reported as a multimodal comparison, not as a replacement for the primary Gemini/GLM experiments.

Table 2: Overall Micro F1 and Macro F1 results on 793 non-Chinese test rows.

System	Micro F1	Macro F1
Gemini 2.5 Pro RAG	**0.6507**	**0.6140**
GLM-5.2 RAG	0.6280	0.5963
GPT-5.4-mini VLM	0.6217	0.5748
MiniLM + LR	0.4603	0.4489

As a lightweight open-source baseline, we train a MiniLM embedding classifier. MiniLM is suitable for this baseline because it compresses Transformer representations through self-attention distillation and is designed for efficient inference [19]. We encode the same SASB-page text representation with `paraphrase-multilingual-MiniLM-L12-v2`, a multilingual sentence-embedding model related to Sentence-BERT and multilingual knowledge distillation [13, 14]. A logistic regression classifier with balanced class weights predicts the three `pred_label` classes; we use it as a transparent linear classification baseline implemented with scikit-learn [20]. Auxiliary `category_match` and `unit_match` fields are derived deterministically from the predicted label for scoring-format compatibility. This baseline is fast and reproducible, but it cannot explicitly reason about completeness, units, or page-level disclosure boundaries.

3.5 Evaluation Metrics

We use Micro F1 and Macro F1 as the main evaluation metrics. Micro F1 aggregates true positives, false positives, and false negatives across all instances before computing the F1 score. It therefore reflects the overall classification performance on the evaluation set. Macro F1 computes the F1 score separately for each class and then averages the class scores. This metric is important for this task because `yes but not complete` is less frequent than `yes` and `no`, but it is central to compliance verification. Reporting both metrics allows us to evaluate both overall correctness and class-balanced performance.

4 Experiments

4.1 Overall Results

Table 2 reports the overall results. Gemini 2.5 Pro achieves the best score, with 0.6507 Micro F1 and 0.6140 Macro F1. GLM-5.2 is close, with 0.6280 Micro F1 and 0.5963 Macro F1. GPT-5.4-mini with a low-detail page image obtains 0.6217 Micro F1 and 0.5748 Macro F1, below the two text-prompt LLM configurations in this run but well above MiniLM. MiniLM with logistic regression is much faster but substantially weaker, with 0.4603 Micro F1 and 0.4489 Macro F1. These results indicate that retrieval-conditioned LLM verification adds meaningful value beyond multilingual semantic embeddings, while a single low-detail image does not automatically improve the decision boundary for partial disclosures.

4.2 Class-Level Results

Table 3 shows class-level F1. The `no` class is strongest for both LLM systems, while `yes but not complete` remains the weakest class. Gemini reaches 0.4835 F1 on `yes but not complete`; GLM-5.2 reaches 0.4717; MiniLM reaches 0.3931. GLM-5.2 has higher recall on the partial class, but lower precision, suggesting that it more often uses the partial label as a cautious middle state. Gemini is more conservative on partial predictions and obtains better overall balance.

Table 3: Per-class F1 scores for the three prediction labels on the non-Chinese test set.

System	yes	ybnc	no
Gemini 2.5 Pro RAG	**0.6327**	**0.4835**	**0.7258**
GLM-5.2 RAG	0.5919	0.4717	0.7250
GPT-5.4-mini VLM	0.6266	0.3686	0.7292
MiniLM + LR	0.5133	0.3925	0.4411

Note: ybnc = `yes but not complete`.

Table 4: Language-level Micro F1 and Macro F1 results for each evaluated non-Chinese language.

System	Metric	English	French	Japanese	Korean	Thai
Gemini 2.5 Pro RAG	Micro F1	0.5473	**0.6771**	**0.7231**	**0.7718**	**0.5537**
	Macro F1	0.4833	**0.6382**	**0.6801**	**0.7467**	0.5129
GLM-5.2 RAG	Micro F1	**0.5473**	0.6406	0.6538	0.7584	**0.5537**
	Macro F1	**0.5090**	0.5971	0.5996	0.7337	**0.5180**
GPT-5.4-mini VLM	Micro F1	0.6269	0.6354	0.6077	0.6711	0.5455
	Macro F1	0.5802	0.5825	0.5555	0.6321	0.4864
MiniLM + LR	Micro F1	0.4726	0.5104	0.4846	0.4430	0.3554
	Macro F1	0.4550	0.4548	0.4767	0.4388	0.3100

4.3 Language-Level Results

Language-level results are shown in Table 4. Gemini is strongest on French, Japanese, Korean, and ties GLM-5.2 on Thai Micro F1. GLM-5.2 obtains the highest English Macro F1, while MiniLM is consistently below the LLM systems, with the largest gap on Thai. GPT-5.4-mini is strongest on Korean among its own language slices (Macro F1 0.6321) and weakest on Thai (0.4864), but remains below Gemini on all five languages in this comparison.

4.4 Cross-Submission Comparison

Table 5 contextualizes our strongest result with the independently reported Subtask 2 results from Cierpa and CSCU. Cierpa reports an overall Macro F1 of 0.637 for its best metric-aware few-shot configuration across six languages [15]. CSCU reports 0.624 Macro F1 and 67.9% accuracy for text-plus-image verification over 967 retained page records [16]. Our Gemini configuration reports 0.614 Macro F1 and 65.07% Micro F1 over 793 non-Chinese rows; the additional GPT-5.4-mini VLM run reports 0.5748 Macro F1 and 62.17% accuracy on the same rows.

These values are close in magnitude but are not directly comparable. Our evaluation excludes Chinese, Cierpa applies a conflict-oriented cleaning procedure, and CSCU reports a retained 967-record set and excludes a documented source-alignment anomaly. The systems also differ in model, prompt, page modality, and example selection. Consequently, Table 5 is a comparison of reported

Table 5: Independently reported best Subtask 2 results. Different row sets and protocols prevent direct ranking.

System	Macro F1	Acc./Micro F1	Scope
Cierpa Best FI	0.637	–	Six languages, cleaned
CSCU text+image	0.624	0.679	967 retained rows
SASBPageRAG Gemini	0.614	0.6507	793, no Chinese
SASBPageRAG GPT-5.4-mini VLM	0.5748	0.6217	793, no Chinese

experimental contexts rather than an official ranking or a paired significance test.

4.5 Result Analysis

The main finding is that the task benefits from explicit LLM-based verification. MiniLM embeddings capture broad multilingual similarity, but the task requires more than semantic closeness. The model must decide whether a page satisfies a specific SASB metric, whether the disclosure is complete, and whether the unit is compatible. This explains why the embedding classifier often fails despite using the same training rows and PDF text.

The comparison between Gemini 2.5 Pro and GLM-5.2 also illustrates that prompt-calibrated retrieval does not remove model-specific decision tendencies. GLM-5.2 predicts `yes but not complete` more frequently, which improves recall for partial disclosures but reduces precision. Gemini 2.5 Pro predicts `no` more often and obtains stronger overall F1 balance. GPT-5.4-mini with a low-detail page image is substantially stronger than MiniLM overall, but its partial-disclosure F1 (0.3686) remains below both text-prompt LLMs, indicating that image access alone does not solve completeness reasoning. In practical compliance screening, the preferred model may depend on whether the user values recall for partial disclosures or conservative final acceptance.

The other submissions refine this interpretation. Cierpa's metric-prioritized few-shot configurations improve its overall Macro F1 from 0.621 for zero-shot inference to 0.634 in cross-validation, and its selected test configuration reaches 0.637. However, the effect is language-dependent: its reported Japanese Macro F1 decreases from 0.748 to 0.663 while Korean increases from 0.671 to 0.832. Thus, examples can calibrate a difficult boundary but can also anchor the verifier to inappropriate edge cases. This observation supports our use of same-metric examples while motivating a controlled comparison against zero-shot and random-example baselines.

CSCU provides complementary evidence about input modality. Its text-only, image-only, and text-plus-image variants report Macro F1 scores of 0.559, 0.605, and 0.624, respectively. The improvement is consistent with our observation that extracted tables lose row and column structure. However, CSCU notes that the confidence intervals for image-only and text-plus-image overlap, so the incremental contribution of extracted text beyond the page image remains uncertain. Taken together, the submissions suggest that demonstration relevance and evidence fidelity address different failure sources: the former calibrates the decision boundary, whereas the latter determines whether the required value, unit, and scope are visible to the verifier.

Cierpa's own end-to-end numbers add a further data point on how these two failure sources compound. When retrieval and label inference are chained, Cierpa's End-to-End Inference Macro F1 falls to 0.287–0.412 across languages, well below its standalone Label Inference range of 0.59–0.83 obtained when the gold page is supplied directly. Cierpa attributes part of this drop to the PDF-level label non-uniqueness discussed in Section 2.5 rather than to retrieval error alone: a correctly retrieved page can still be scored against a mismatched gold label when a report legitimately contains both a `yes` and a `no` page for the same metric. This is a useful caution for our own Section 4.7: any future attempt to chain retrieval into SASBPageRAG's verifier should expect a similar, partly label-driven drop rather than attributing it solely to retrieval quality.

4.6 Error Analysis

Errors remain concentrated around partial disclosure. A page may contain relevant ESG language but omit a required denominator, scope, disaggregation, event count, or unit. Conversely, a page may present a SASB index that names a metric but does not contain the underlying disclosure. Single-page evaluation makes these cases difficult because evidence may appear on neighboring pages. The current system also relies on extracted PDF text; tables or figures that are poorly extracted may be underused.

The observed errors can be summarized into three common patterns. First, topical matches are sometimes confused with compliant disclosures: the page discusses the correct ESG issue but does not provide the metric-specific quantity, category, or unit. Second, partial disclosures are hard to separate from complete disclosures when the page provides proxy evidence, percentages without denominators, or qualitative descriptions without the required numerical detail. Third, extracted text from PDF tables may lose row or column structure, making it difficult for the verifier to connect a number with the correct SASB metric.

These patterns recur across submissions. Cierpa reports that few-shot examples substantially improve the Korean partial class but reduce Japanese partial-label accuracy, indicating that the `yes but not complete` boundary is sensitive to demonstration choice. CSCU likewise identifies this label as the hardest class and attributes remaining errors to incomplete page evidence and confusion between full and partial disclosure. Its Subtask 1 analysis additionally shows that many retrieval errors fall on a page adjacent to the annotated page: for its VLM-Guided variant, Near@1 (46.6%) is roughly one and a half times the exact-page macro F1 (31.0%), a gap the Lexical Reference shows as well (10.4% versus 14.8%). Although our system receives the official target page and does not evaluate retrieval, this finding exposes a limitation of strict single-page verification: the verifier can be penalized for missing evidence that is structurally associated with the target disclosure but printed on a neighboring page. A consolidated, cross-method error taxonomy and additional worked examples are maintained alongside this report in `docs/CLAUDE_ERROR_ANALYSIS.md`.

4.7 Task 1 Retrieval Extension

The RegCom shared task also defines Subtask 1 as full-report compliance matching: a system must localize relevant pages and then verify the disclosure category and unit. The experiments above do not evaluate that setting because Subtask 2 supplies the task-provided page directly. We therefore reconstruct report–metric query groups from the available page-level files and evaluate the page-localization stage separately.[1] Since the official Subtask 1 input records, gold page sets, empty/no-disclosure rules, and evaluator are not included in this checkout, the scores below should be interpreted as reconstructed retrieval measurements rather than official leaderboard values.

The retrieval pipeline follows the general shape of Cierpa's retrieval design (Section 2.5): independent lexical and dense candidate generators, weighted reciprocal-rank fusion, a lane-union candidate pool, and Top-10 vision-language-model (VLM) reranking, evaluated against the reconstructed query groups. On 490 report–metric groups (369 with at least one non-empty gold page, 121 without), the frozen dense-fused retrieval configuration reaches Hit@1 0.220, Hit@5 0.493, Hit@10/candidate recall 0.631, and MRR 0.350 on the non-empty groups. Full `gpt-5.4-nano` reranking of the same Top-10 candidates raises these to Hit@1 0.328, Hit@5 0.583, and MRR 0.431, while Hit@10 remains 0.631 because reranking cannot recover a page absent from the candidate pool. Candidate recall at 10 varies sharply by language (English 0.891, French 0.840, Korean 0.686, Chinese 0.576, Japanese 0.545, Thai 0.313), which identifies Thai, Chinese, and Japanese candidate generation, rather than reranking quality, as the current bottleneck – the same error-propagation pattern CSCU reports for its own retrieval design. The 490 VLM traces completed successfully after rate-limit retries, using 6,758,052 input and 143,304 output tokens at an estimated cost of US$1.531. A separate cached run feeds the VLM-selected Top-1 page and page text to a lightweight verifier, obtaining a predicted-label distribution of 301 `no`, 143 `yes but not complete`, and 46 `yes`; only 74 selected pages join unambiguously to an available truth row once the label non-uniqueness discussed in Section 4.6 is taken into account, and label accuracy on that 74-row subset is 0.459.

A deeper-candidate follow-up configuration (`task1-neighbor2`: the same dense-enabled fusion, plus a two-page neighbor expansion around the Top-10 anchors, doubling the VLM candidate pool to Top-20) was completed on all 490 groups. The candidate-pool recall rises to 0.710 at Top-20 (0.789 at Top-30), but the 20-image nano reranker reaches Hit@1 0.314, Hit@5 0.561, Hit@10 0.648, Near@1 0.398, and MRR 0.423, compared with Hit@1/5/10 0.328/0.583/0.631 and MRR 0.431 for the 10-image primary configuration. The larger pool therefore improves coverage at Hit@10 but slightly reduces Top-1 and MRR, suggesting that the small model becomes less decisive when shown more visually similar pages. The effect is language-dependent rather than uniform: Chinese, Japanese, Korean, and Thai Top-1 all improve with the larger pool (e.g. Japanese Hit@1 0.182 to 0.309), but French Top-1 accuracy *regresses* (0.320 to 0.300, MRR 0.484 to 0.467) even though French candidate recall also improves (0.840 to 0.900) – the same qualitative pattern Cierpa reports for adding few-shot context by language (Section 2.5), here obtained by varying candidate-pool size instead. Consequently neither configuration strictly dominates the other, and a single frozen configuration for future reporting remains an open choice rather than an automatic one. The complete run used 13,178,046 input and 275,396 output tokens at an estimated cost of US$2.980.

The completed local BGE-M3 ablation uses the same reconstructed query groups and page mapping and runs without API calls. It compares dense retrieval, dense+sparse fusion, dense+sparse+ColBERT multi-vector fusion, and BGE-Reranker-v2-M3 reranking. Table 6 reports the unified scores on the 369 query groups with non-empty gold pages; the full reconstructed input contains 490 groups, including 121 empty-gold groups.

The best first-stage configuration is the three-representation BGE-M3 fusion, improving over the MiniLM baseline by 4.6, 5.2, and 4.4 percentage points on Hit@1, Hit@5, and Hit@10, respectively, and by 0.048 MRR. The gains are consistent with the component ablation: sparse features add about one point over dense-only retrieval, and the ColBERT-style multi-vector lane adds a further 1.5–2 points. Natural-language query rewrites underperform the original metric-keyword queries (0.209–0.228 Hit@1 across the tested variants), suggesting that the added English "what counts" wording introduces noise for non-English reports. Increasing the BGE maximum length from 2,048 to 8,192 tokens has almost no effect because most page texts are shorter than the limit. In contrast, the BGE reranker lowers Top-1 and MRR (0.160 and 0.313): it often promotes SASB/GRI index pages that mention a metric code over the page containing the actual disclosure. Language-specific results show the largest Top-1 gains for Korean (0.216 to 0.373) and Japanese (0.182 to 0.236), while Thai remains the principal candidate-generation bottleneck (Hit@10 0.281). These findings support stronger multilingual candidate generation, but also show that reranking quality and disclosure-page relevance must be analyzed separately. A 65-row stratified, human-review-only error sample is provided in `docs/TASK1_ERROR_REVIEW_SAMPLE.csv`; its qualitative judgment columns intentionally remain blank for manual annotation.

To quantify uncertainty, we used 10,000 paired bootstrap resamples over the same 369 non-empty groups. The best BGE fusion improves Hit@1, Hit@5, Hit@20, and MRR significantly relative to the baseline, while the positive Hit@10 and Near@1 differences include zero in their 95% intervals. A paired McNemar test on Hit@1 gives 22 baseline-only versus 39 BGE-only successes ($p = 0.040$). The raw-versus-natural query comparison is significant for the multi-vector system (22 versus 8, $p = 0.016$), and the all-language reranker significantly degrades Top-1 (58 versus 25, $p = 0.0004$); language gating recovers a significant portion of those errors (14 versus 48, $p < 0.001$).

The error analysis separates candidate recall from ranking. The reranker selects a SASB/GRI index page as Top-1 for 91/369 raw queries (66 are incorrect) and 142/369 natural queries (111 incorrect). Representative failures include an esun query whose gold page 14 is replaced by a GRI index at page 195, and two hyundai metrics whose gold pages 72–73 are both replaced by the same SASB index at page 114. Thai and French fail for different reasons: Thai has only 0.281 Hit@10 and misses the gold page from the Top-20 pool for 32/64 groups, whereas French reaches 0.860 Hit@10 and its main problem is ordering (29/50 gold pages are in Top-20 but not Top-1). The 121 empty-gold groups also expose a limitation of the current interface: the retriever always returns candidates, so abstention detects 0/121 cases, and the fused Top-1 score has AUC 0.48 for distinguishing empty from non-empty groups. Finally, three out-of-range gold pages in the SPT report are caused by a one-page extraction failure; because every arm uses the same page map, they affect comparisons equally.

[1] Full run configurations, dates, and the claim-boundary policy are maintained in `docs/EXPERIMENT_LOG.md` and `docs/TASK1_VLM_PLAN.md` in the project repository.

Table 6: Task 1 page-retrieval ablation on the reconstructed set (369 non-empty-gold groups).

Configuration	Hit@1	Hit@5	Hit@10	Hit@20	Near@1	MRR
MiniLM fused baseline	0.220	0.493	0.631	0.751	0.293	0.350
BGE-M3 dense	0.238	0.504	0.631	0.748	0.290	0.369
BGE-M3 dense+sparse	0.249	0.523	0.648	0.772	0.312	0.381
BGE-M3 dense+sparse+ColBERT	**0.266**	**0.545**	**0.675**	**0.818**	**0.325**	**0.398**
BGE-Reranker-v2-M3 (raw, all languages)	0.160	0.504	0.672	–	–	0.313

Table 7: Self-reported strengths and weaknesses of the three RegCom submissions compared in this report.

System	Strengths	Weaknesses
SASBPage-RAG	Transparent TF-IDF and BGE-M3 retrieval; label-balanced examples; optional GPT-5.4-mini page-image verifier; reproducible and modular	Single-page image is low-detail and loses cross-page context; multimodal run is below Gemini on Macro F1; no zero-shot or random-example control; non-Chinese scope only
Cierpa	Multi-signal retrieval (lexical, semantic, visual) with RRF; explicit dataset non-uniqueness audit; systematic few-shot ablation (ZI/FI/xFI × random/metric-prioritized)	Trained fusion weights (OptRRF) underperform untuned fusion on test; VLM reranking cost is high (US$11–$62 per 615 instances); few-shot gains are language-dependent and can reverse (Japanese)
CSCU	Single-factor modality ablation (text/image/text+image); bootstrap CIs and paired significance tests; explicit seen-document, unseen-query framing	VLM-Guided variant changes section selection and final verification jointly, so the gain is not attributable to one component; Subtask 1 and 2 evaluated independently, not end-to-end; exact-page scoring penalizes adjacent-page evidence

5 Discussion

5.1 Strengths and Weaknesses across Approaches

Table 7 summarizes the strengths and weaknesses each submission reports for itself. The three approaches occupy different points in the accuracy–complexity space. SASBPageRAG now has a modular text-first path plus optional BGE-M3 retrieval and GPT-5.4-mini page-image verification. This keeps the core pipeline easy to inspect and comparatively inexpensive, while allowing visual evidence for pages whose layout is poorly represented by extracted text. Its central weakness is that the optional multimodal path still uses a single low-detail image and does not fully preserve cross-page context.

Cierpa offers the broadest retrieval design. Lexical, semantic, and visual rankings capture complementary signals, and VLM reranking raises its reported retrieval MRR from 0.497 without reranking to 0.584 with GPT-5.4-mini and 0.636 with GPT-5.5. The cost is a larger pipeline with multiple representations and model calls; Cierpa reports reranking costs of US$11.0 and US$62.1, respectively, for 615 test instances. Its few-shot study is particularly informative for our setting because it shows that metric alignment is useful overall but not uniformly beneficial across languages.

CSCU contributes a strongly traceable design and an explicit modality comparison. Its results show that whole-page visual evidence can recover information lost by text extraction, while its retrieval results distinguish candidate-generation failures from verifier failures. A limitation is that its VLM-guided retrieval variant changes ToC checking, section selection, and final verification together; the reported improvement therefore measures a package of interventions rather than an isolated ablation. Its use of remote multimodal services also creates cost and model-drift concerns shared by other API-based systems.

5.2 Evaluation Scope and Validity

Macro F1 is more informative than accuracy alone because the partial class is less frequent and consistently difficult. Nevertheless, a single aggregate score hides material differences in language coverage, class distribution, evidence quality, and retained records. Cross-submission scores should therefore be accompanied by their evaluation scope, as in Table 5, and should not be compared as if generated on an identical paired sample.

The released reference data also complicate evaluation. Cierpa's own audit of the official release quantifies this: across all six languages, 21.1% of training and 16.5% of test relevant-page targets are non-unique for the same metric tuple, and 55.6% of training and 57.2% of test instances share a non-unique gold label at the PDF level, because a `no` label marks absence from one page rather than from the whole report. An evaluator that joins rows only by a non-unique tuple can collapse instances and change the reported result. Reviewer-facing analysis should preserve row-level identity or introduce a unique sample identifier. Separately, CSCU's audit of the official Subtask 1 split found that all official test documents also appear in the training split, which it reports as seen-document, unseen-query evaluation rather than unseen-document generalization; because Cierpa, CSCU, and SASBPageRAG all draw on the same official RegCom release, this framing plausibly applies to the other submissions' splits as well, though we have not independently confirmed it for the non-Chinese CSV split used here. Moreover, the available reference files do not provide independently validated gold labels for the predicted `category_match` and `unit_match` fields; these auxiliary outputs should not be scored by treating an answer-unit string as an equivalent categorical gold label.

Finally, exact single-page evaluation is intentionally controlled but does not represent every real report. Tables may continue across pages, narrative definitions may precede a numerical table, and SASB index pages may point to rather than contain evidence. Future evaluation could complement exact-page labels with acceptable page sets, neighboring-page sensitivity, or short evidence-span annotations. These alternatives should supplement rather than silently replace the official protocol.

5.3 Implications for System Design

The comparison suggests a modular research direction. Metric-prioritized examples can improve decision calibration, while page images can improve evidence fidelity; neither substitutes for the other. A practical extension of SASBPageRAG would preserve its lightweight text path for narrative pages and route table-heavy, scanned, or low-text pages to a multimodal verifier. A controlled ablation among zero-shot, random few-shot, TF-IDF few-shot, and metric-prioritized few-shot settings would then isolate the contribution of retrieval separately from the underlying LLM. Such experiments are preferable to attributing all gains to "RAG" when the model, prompt, sample scope, and evidence modality also differ.

5.4 Limitations

Our conclusions are subject to several limitations. First, our primary results cover 793 non-Chinese rows and therefore do not constitute a complete six-language evaluation. Second, Gemini 2.5 Pro and GLM-5.2 are remote models whose behavior and availability may change, while exact generation-level seeds and provider-side versions were not preserved. Third, our current comparison includes two LLMs and one embedding classifier but does not isolate the effect of retrieved demonstrations against zero-shot and random-example controls. Fourth, page evidence is extracted as text, so tables, figures, reading order, and scanned content may be degraded. Fifth, raw scores from the other submissions are contextual evidence rather than paired comparisons because their cleaning, retained samples, and modalities differ; unlike CSCU, neither Cierpa nor this report currently attaches bootstrap confidence intervals or paired significance tests to its headline numbers. Sixth, the preliminary Task 1 retrieval extension in Section 4.7 is diagnostic only: the official evaluator and gold page-set contract are not yet confirmed, so none of those numbers should be read as official. Finally, our qualitative error taxonomy identifies plausible mechanisms but does not yet provide a fully annotated error sample with inter-annotator agreement.

6 Conclusions

This paper presented SASBPageRAG, a retrieval-augmented few-shot verifier for multilingual single-page SASB metric verification. The main research contribution is a lightweight verification framework that combines TF-IDF case retrieval, label-balanced few-shot prompting, SASB metadata, and extracted page evidence without fine-tuning a language model. The experimental results show that retrieval-conditioned LLM verification is more effective than a MiniLM logistic-regression baseline for judging whether a single ESG report page satisfies a metric, category, and unit requirement.

The comparative analysis clarifies that no single component resolves the task. Our experiments show the value of explicit LLM verification over embedding classification; Cierpa demonstrates the potential and language dependence of metric-prioritized examples; and CSCU demonstrates the value of visual page evidence. Across all three submissions, partial disclosure remains the central challenge, especially when relevant ESG language appears without the required quantity, denominator, scope, disaggregation, or unit. ESG compliance verification should therefore be studied as both a metric-level evidence-matching problem and a calibrated completeness decision, rather than only as document-level semantic classification.

The practical implication is that a lightweight text path can handle many pages while costlier multimodal reasoning is reserved for low-text or table-heavy cases. The completed BGE-M3 ablation shows that multi-vector candidate generation is a useful no-API upgrade, but that a generic reranker can promote index pages and hurt Top-1 ranking. Future work will isolate zero-shot, random-example, and metric-prioritized retrieval effects; add OCR and page-image evidence under a controlled modality ablation; and study neighboring-page context separately from the single-page setting. Row-level identifiers, paired statistical tests, and explicit reporting of retained samples will also be important for reliable comparison.

Resource Availability

The supplementary resources for this report, including the source code, public reproducibility inputs, evaluation scripts, and model outputs/predictions, are available at: https://github.com/wesleyseawatch2/sasb-page-rag-regcom and https://drive.google.com/drive/folders/1z5DGv2I4TL9sw9BCBBohASZp3HHLj5Uo?usp=sharing.

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